

ENVISION2020

Zemax

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Advanced ZPL Programming

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Example 1: Create your own analysis

To implement features or calculations not in the program...

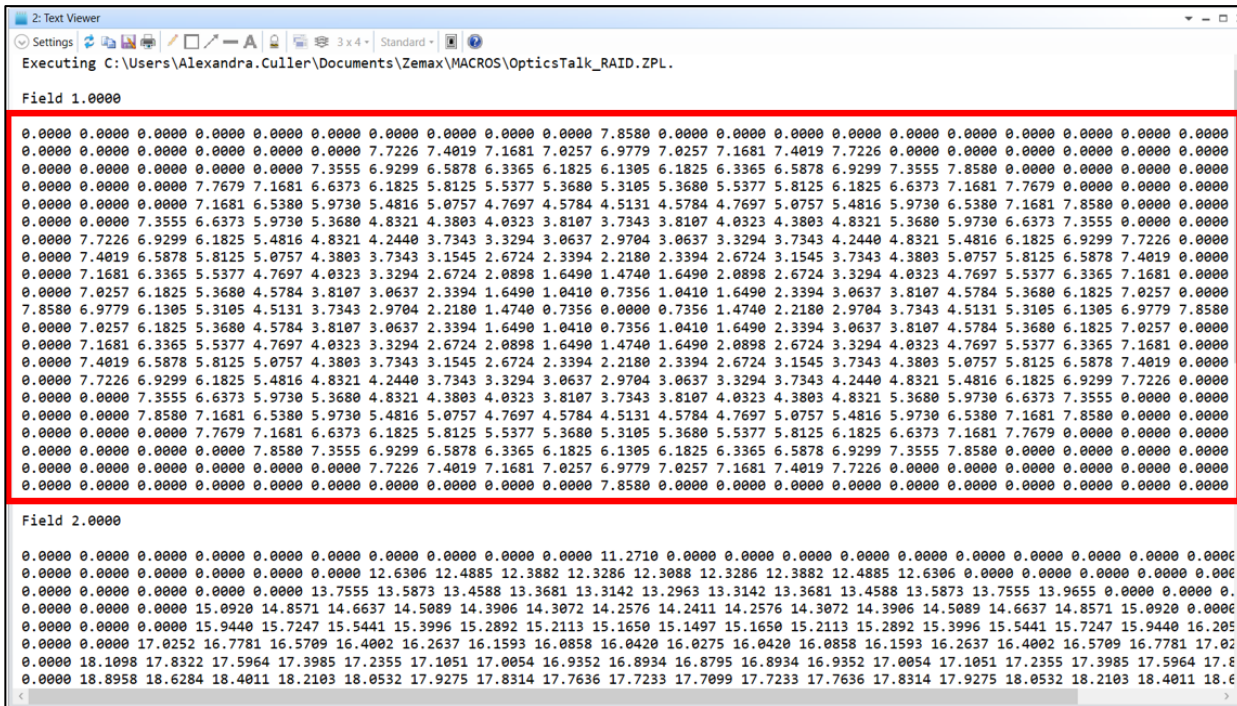
- Example: Calculate the angle of incidence (AOI) for all rays across the pupil at a given surface
 - Can be used to check for high AOI which would indicate a very sensitive optic (difficult to manufacture at low cost)
- What we need:
 - A method to calculate AOI
 - A way to calculate the AOI for multiple rays
- The optimization operand RAID will calculate the Angle of Incidence for a particular field and pupil coordinate, at a given surface, for a specific wavelength

	Type	Surf	Wave	Hx	Hy	Px	Py		Target	Weight	Value	% Contrib
1	RAID ▾	5	2	0.00000000	1.00000000	0.40000000	-0.40000000		0.00000000	0.00000000	26.61496516	0.00000000

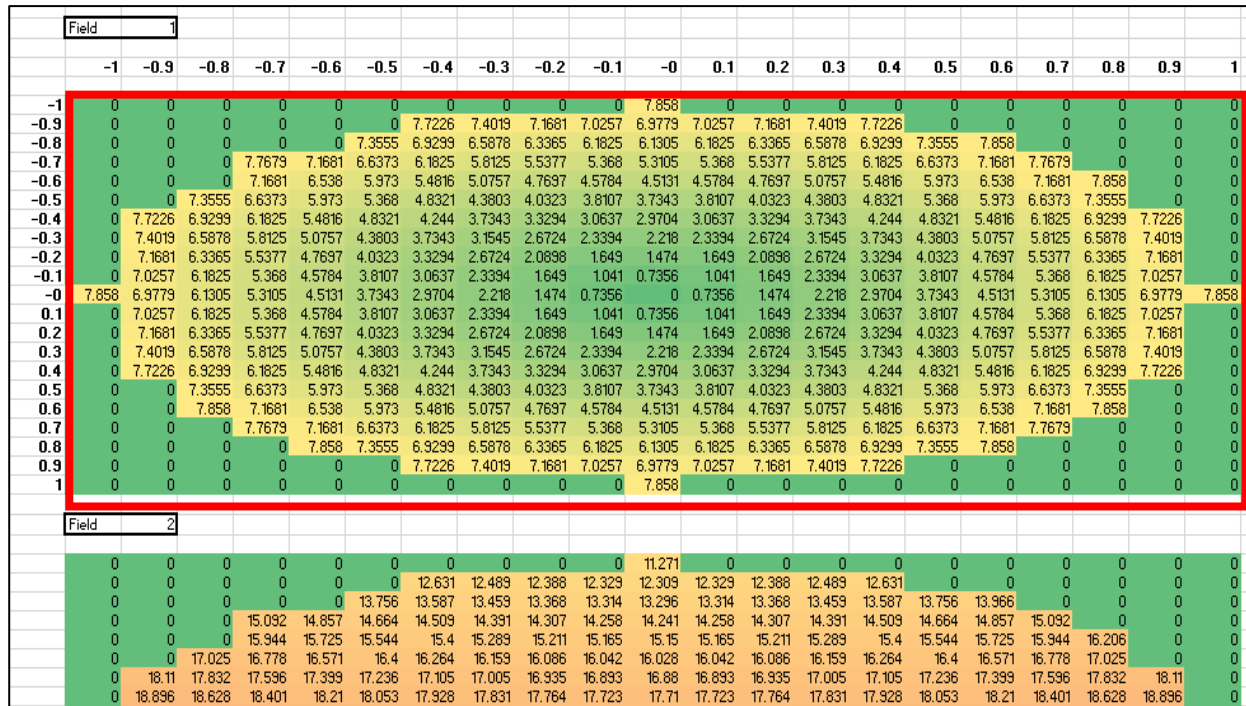
AOI analysis: Procedure

1. Choose the surface, number of rays to trace in X and Y across the pupil, and wavelength of interest.
2. To assess across all fields, declare an array to hold the field data.
3. Find the maximum field in the system for calculation of H_x and H_y .
4. Use a nested FOR loop to cycle through fields (1), X-direction (2), Y-direction (3).
5. Use optimization operand RAID to report the incident angle.

AOI analysis: Results



OpticStudio output



Post-processing results

Can we clean this up?

- Instead of hard-coded inputs, ask the user for the surface, step number, and wavenumber of interest.
- Instead of a nested FOR loop, use CALLMACRO
 - CALLMACRO gives us the ability to call a second (child) macro from within the first (parent)
 - CALLMACRO is analogous to calling an external function file. We typically save common tasks to function files for use later.
- Data may be passed back and forth between the parent and child macros using the keywords CALLSETDBL and CALLSETSTR and the functions CALD and \$CALLSTR
- More information:
 - [The Programming Tab](#) > [About the ZPL](#) > **Calling a Macro from within a Macro**
 - MyZemax Knowledgebase: "How to call a ZPL macro from within a ZPL macro"

Update the macro

- Global variables (to be passed to the child macro):
 - Surface number, wave number, number of rays (numSteps), hx, hy, and the operand of interest
- Child macro will now be used to walk across the pupil for any operand which has the same properties as RAID:

Type	Surf	Wave	Hx	Hy	Px	Py	Target	Weight	Value	% Contrib
------	------	------	----	----	----	----	--------	--------	-------	-----------


```
OpticsTalk_RAID_Parent.ZPL
33  hy = yFlds(i)/maxFldY
34  ENDIF
35  IF (maxFldX != 0)
36    hx = xFlds(i)/maxFldX
37  ENDIF
38  PRINT
39  PRINT "Field ", i
40  PRINT
41
42  operand = OCOD("RAID")
43
44  CALLSETDBL 1, numSteps
45  CALLSETDBL 2, surf
46  CALLSETDBL 3, wave
47  CALLSETDBL 4, hx
48  CALLSETDBL 5, hy
49  CALLSETDBL 6, operand
50  CALLMACRO OpticsTalk_RAID_Child.ZPL
51
52 NEXT
```

```
OpticsTalk_RAID_Child.ZPL
numSteps = CALD(1)
surf = CALD(2)
wave = CALD(3)
hx = CALD(4)
hy = CALD(5)
operand = CALD(6)

DECLARE rayData, double, 2, numSteps, numSteps

! Walk across pupil for each field
FOR j, -1, 1, 1/numSteps
  iterX = 1
  px = j
  FOR k, -1, 1, 1/numSteps
    iterY = 1
    py = k
    IF ((px*px) + (py*py) > 1)
      rayData(iterX, iterY) = 0
    ELSE
      IF ((px*px) + (py*py) <= 1) THEN rayData(iterX, iterY) = 0
```


Other potential updates:

- Use OUTPUT to print directly to an external CSV file. Use \$TAB() as the delimiter
- Use this data to find maximum AOI. To do so, store the results in an array using DECLARE